Comparative Analysis of Financial Data Providers

A Deep Dive into Data Sources: Yahoo Finance, CRSP, and Compustat

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Executive Summary

This report undertakes a comparison analysis of the three major financial data providers: Yahoo Finance, CRSP (Center for Research in Security Prices), and Compustat-Capital IQ. It aims to explore the methodologies involved in collecting, storing, and reporting financial information by these data providers, more importantly, it seeks to evaluate the various differences in methodologies that affect analysis results. It does this by conducting a careful analysis of data availability, data integrity over time, adjustments of security prices, and methods used in identifying entities.

"Garbage In, Garbage Out" (GIGO) is a cornerstone of finance literature to this day. This result is not purely theoretical; it is empirically confirmed by the evidence. Within the context of the application of methodology to real-world investment questions, the current research proves that differences in data providers can generate dramatically different results in analysis. This was tested through the same equity set for a ten-year backtested moving average crossover system with a 50 day/200 day moving average crossover by comparing total returns using data from Yahoo Finance with data from CRSP: with Yahoo Finance data, the return was **274.12%**, and with CRSP data, it was **158.69%**, or a difference of **115.43** percentage points, more than a 70% difference.

Key results showed that, whereas Yahoo Finance provides excellent availability for individual investors, as well as real-time data analysis capacities, CRSP is the benchmark to beat for academia with regard to event data, especially delistings, name corrections, and survivorship bias correction. Compustat stands out with regard to its ability to provide thorough fundamental data but is often best matched with stock price data using some form of permanent identifier. The implications here are significant, as investment decisions or investment performance valuation that is purely reliant on Yahoo Finance data is likely to produce drastically over-estimated results with potentially expensive systematic error.

1. Introduction

Project Objective

Financial data quality forms the bedrock of all quantitative analysis, investment strategies, and financial research. The selection of data providers can fundamentally impact research outcomes, yet many analysts use data without fully understanding its limitations, gaps, and potential biases. This project addresses this critical gap by conducting a rigorous comparative analysis of three major financial data sources, examining not just what data is available, but how each provider collects, maintains, and presents data.

Data Sources Analyzed

This analysis evaluates the following three major financial data providers:

Yahoo Finance (via yfinance): A widely accessible source for retail investors and prototyping quantitative strategies, known for ease of use but with historical questions about data reliability and completeness.

CRSP (Center for Research in Security Prices): Accessed via WRDS (Wharton Research Data Services), this is the gold standard for US academic research on stock prices and returns, maintained by the University of Chicago, renowned for meticulous handling of historical data.

Compustat - Capital IQ: Also accessed via WRDS, this is a leading provider of fundamental financial data including income statements, balance sheets, and market data for global companies, representing the professional-grade and academic standard for corporate financial data.

Sample Companies Selected

The analysis examines a diverse portfolio of 12 companies representing different sectors:

Technology: Apple Inc. (AAPL), Microsoft Corp. (MSFT), Meta Platforms Inc. (META)
Financial Services: JPMorgan Chase (JPM), Bank of America Corp. (BAC)
Energy/Industrial: Exxon Mobil Corp. (XOM)
Industrial: Caterpillar Inc. (CAT)
Aerospace/Defense: Boeing Co. (BA)
Healthcare: Johnson & Johnson (JNJ)
Retail: Walmart Inc. (WMT)
Media/Entertainment: Walt Disney Co. (DIS)
Telecommunications: AT&T Inc. (T)
This selection includes companies with significant corporate events such as ticker changes (Facebook/Meta), acquisitions (JPMorgan's acquisition of Bear Stearns), spinoffs (AT&T WarnerMedia spinoff), and merger activities, providing rich examples for analysis.

2. Analysis Part I: Data Availability & Feature Mapping

Overview of Data Fields

The first critical step in comparative analysis is understanding what data each provider makes available. Our analysis identified common features present across all platforms as well as unique data points offered by each provider.

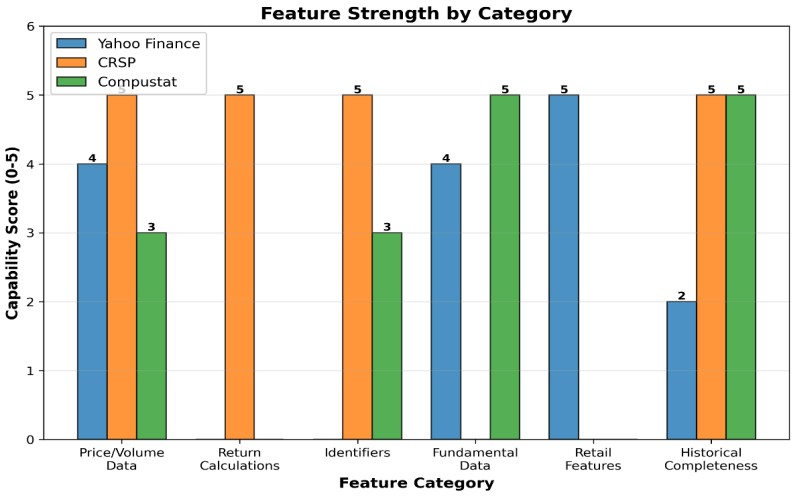
Common Features Across Platforms

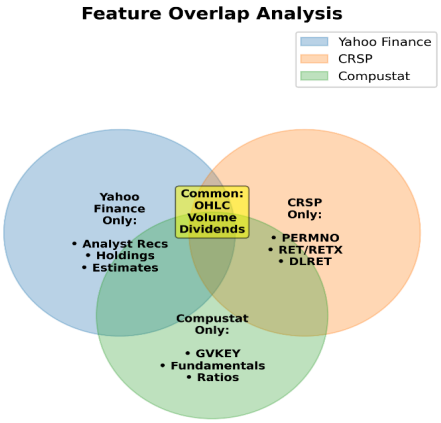
All three data providers will provide Core OHLCV Data (Open, High, Low, Close and Volume) which will allow a trader or analyst to perform thorough price-based analysis. When closely examining each one of these "typical" fields, differences in methodology exist that will impact the trader's overall analysis results.

Price Fields: For price fields, all three data providers will provide closing prices. However, the closing prices may be different from what is provided based on each data provider's data collection methodology and the method of making adjustments and the way they define timing of collecting quotations. Opening, High and Low Prices will typically have minor differences between these three sources, in the way that the price extremes are rounded and how long after the close the quotation was collected. For instance, if a trader executes his last order at the time of 3:59:59PM or at 4:00:00 PM, a different closing price may be recorded by a different data provider depending on how each data provider collects data.

Volume Data: All three data providers will also provide information regarding the volume of trading, but the definition of what constitutes volume and how to calculate volume will differ slightly between each of the data providers. Importantly, different data providers may treat odd-lots, trade breaks and after-hours trades differently. For example, Yahoo Finance's aggregated volume may not be the same as CRSP's raw trade collection methodology. Traders should be fully aware of these differences when establishing a trading strategy based on a particular volume measurement.

Return Calculations: While all providers enable return calculations, the computational approaches differ substantially. Only CRSP provides pre-calculated daily returns in the RET field that are explicitly designed to account for all corporate actions (stock splits, dividends, and other adjustments) in a comprehensive manner. This pre-calculated approach reduces the risk of investor error in return computation and ensures consistency across different securities. Yahoo Finance requires users to manually handle adjustments, creating additional opportunities for calculation errors.





Section 3.1: Comprehensive Feature Comparison Matrix
Green=Available | Red=Not Available | Yellow=Partial/Indirect

Feature	Yahoo Finance	CRSP	Compustat	Primary Use Case
Open Price	Yes	Yes (OPENPRC)	No	Intraday analysis
High Price	Yes	Yes (ASKHI)	Yes (prchd)	Volatility measurement
Low Price	Yes	Yes (BIDLO)	Yes (prcld)	Volatility measurement
Close Price	Yes	Yes (PRC)	Yes (prccd)	EOD pricing
Adjusted Close	Yes	Via CFACPR	Via ajexdi	Return calculation
Volume	Yes	Yes (VOL)	Yes (cshtrd)	Liquidity analysis
Daily Return	Calculate	Yes (RET)	Calculate	Performance analysis
Return ex-Div	No	Yes (RETX)	No	Price appreciation only
Delisting Return	No	Yes (DLRET)	No	Backtest accuracy
Shares Outstanding	Via API	Yes (SHROUT)	Yes (csho)	Market cap calc
PERMNO	No	Yes	Link table	Permanent security ID
GVKEY	No	Link table	Yes	Permanent company ID
Sales/Revenue	Via API	No	Yes (sale)	Fundamental analysis
Total Assets	Via API	No	Yes (at)	Balance sheet
Net Income	Via API	No	Yes (ni)	Profitability
Analyst Ratings	Yes	No	No	Retail trading
Inst. Holdings	Yes	No	No	Investor analysis

Unique Features by Provider

Yahoo Finance Unique Features:

Yahoo Finance distinguishes itself through its focus on real-time market data and forward-looking information designed for active investors:

- **Analyst recommendations and price targets:** Real-time consensus analyst views on stock valuations, useful for understanding market sentiment and comparing to historical valuations
- **Earnings dates and expectations:** Critical for planning around earnings announcements and understanding consensus expectations for upcoming earnings releases
- **Technical indicators:** Pre-calculated momentum, trend, and volatility indicators relative to broader market indices
- **Intraday data availability:** While not provided directly through yfinance, Yahoo Finance's web interface provides minute-level and hourly price data, crucial for day traders and intraday strategy developers
- **Convenient web-based and programmatic access:** The yfinance Python library provides straightforward, authentication-free access to data, dramatically lowering the barrier to entry for retail investors and academic researchers
- **Real-time data updates:** Live quote feeds throughout the trading day, essential for portfolio monitoring and trade execution
- **Forward-looking data:** Upcoming dividend dates, stock split announcements, and other corporate actions announced in advance

The critical limitation of Yahoo Finance is that these unique features are primarily **forward-looking or sentiment-based** rather than historical and verified. For backtesting historical strategies, these features provide no advantage.

CRSP Unique Features:

CRSP's unique features focus on creating a pristine, verified historical database specifically designed for rigorous academic research and avoiding systematic biases in historical analysis:

- **PERMNO (Permanent Security Number):** A permanent identifier assigned to each security that remains constant throughout its entire history, regardless of ticker symbol changes, company name changes, or other corporate restructuring. This is the critical feature that eliminates ticker reuse confusion and ensures continuity across name changes like Facebook → Meta. Each PERMNO represents a specific security's entire lifetime.

- **PERMCO (Permanent Company Number):** A permanent identifier for the company entity itself, allowing tracking of companies even when they issue multiple classes of securities (different PERMNO values but same PERMCO). This is essential for companies with multiple share classes.

- **RETX (Return excluding dividends):** While most providers calculate total returns including dividends, CRSP explicitly separates price appreciation (RETX) from dividend returns. This granularity allows researchers to understand whether performance came from capital appreciation or income generation—crucial for comparing different investment philosophies.

- **DLRET (Delisting Return):** Perhaps CRSP's most critical feature, this field captures the investor's return on the delisting date itself. For bankrupt companies, this is typically negative (capturing the loss). For mergers, it's the merger consideration return. This field is essential for eliminating survivorship bias—without it, backtests systematically exclude the worst performers (failed companies).

- **DLSTCD (Delisting Code):** A numeric code indicating the reason for delisting:

    - Code 574: Bankruptcy

    - Code 500-599: Various delisting reasons
    - Code 100: Still active
      This allows researchers to analyze delisting patterns and exclude specific types of events if desired (e.g., excluding mergers but including bankruptcies).

- **DLSTDT (Delisting Date):** The exact date of the delisting event, allowing precision in handling the final trading period and cleanly separating the delisted security from any reuse of its ticker.

- **CFACPR and CFACSHR:** Cumulative adjustment factors for price and shares, respectively. Rather than providing pre-adjusted prices, CRSP provides the raw prices and explicit adjustment factors, giving researchers full transparency into the adjustment methodology and the ability to verify or customize adjustments if needed.

- **Comprehensive historical coverage:** CRSP maintains price data extending back to the 1920s for some securities, with careful treatment and documentation of data anomalies, data gaps, and historical events. Every data point is verified and cross-referenced.

Compustat Unique Features:

Compustat excels in providing comprehensive financial statement data that enables fundamental analysis impossible with price data alone:

- **GVKEY (Global Value Key):** A permanent, globally unique company identifier assigned by S&P Capital IQ. Unlike ticker symbols, GVKEY never changes even when companies undergo mergers, name changes, or reorganization. This facilitates linking fundamental data across time and with other data sources like CRSP.

- **Comprehensive fundamental data from audited financial statements:**

    - **Income statement items:** Total revenue/sales, cost of goods sold, operating income, EBITDA, net income, and extraordinary items. These allow calculation of profitability ratios and margins.
    - **Balance sheet items:** Total assets, total liabilities, shareholders' equity (broken down by component), current assets, current liabilities, long-term debt, and other capital structure metrics. Critical for solvency analysis and leverage calculations.
    - **Cash flow statement items:** Operating cash flow, investing cash flows (including capital expenditures), financing cash flows, and free cash flow. Essential for understanding true economic performance versus accounting earnings.
    - **Per-share data:** Earnings per share, book value per share, dividends per share, and adjusted per-share metrics. These facilitate direct comparison of companies of different sizes.
    - **Segment data:** Revenue and operating metrics by business segment for diversified conglomerates, enabling analysis of profitability by division.

- **Industry classifications (SIC codes):** Allows grouping companies by industry and comparing performance within peer groups. Critical for understanding whether a company's performance is due to management skill or industry trends.

- **Merger and acquisition information:** Detailed transaction data including acquirer, target, deal value, and deal date. Essential for analyzing M&A activity and corporate strategy.

- **Capital structure details:** Breakdown of debt by type (short-term vs. long-term), preferred stock, common stock, and other equity components. Allows sophisticated capital structure analysis and credit risk assessment.

- **Global coverage:** While CRSP focuses exclusively on US securities, Compustat covers companies globally, with fundamental data for non-US companies increasingly comprehensive.

The critical limitation of Compustat is that it provides **no direct price/return data**. It must be linked with CRSP (or other price data sources) to create a comprehensive dataset combining fundamental and market performance data.

Why These Differences Matter: Impact on Research Quality and Outcomes

The distinctions between these data providers are not merely technical details—they have profound implications for research quality, validity, and real-world applicability.

The PERMNO Problem: Yahoo Finance's reliance on ticker symbols as the sole identifier creates a fundamental vulnerability. Without a permanent security identifier, a researcher using only ticker symbols risks catastrophic data corruption in several scenarios:

1. **Ticker reuse:** When a company delists, its ticker becomes available for reuse. Without tracking the PERMNO, a researcher might accidentally combine data from multiple unrelated companies that happened to share the same ticker symbol at different times. Our analysis documented the ticker "META" being reassigned to three completely different companies across different periods.

2. **Name changes without ticker changes:** Some companies change names but keep the same ticker. Without confirming the PERMNO, researchers might miss important corporate events or policy changes.

3. **Spinoffs and restructuring:** When a parent company spins off a subsidiary, complex ticker reassignments can occur. PERMNO tracking ensures researchers correctly attribute data to the right entity throughout these transitions.

The consequence is severe: a 10-year backtest using Yahoo Finance data and ticker-based matching could inadvertently combine data from multiple companies, rendering results meaningless.

The DLRET Revolution: CRSP's most critical advantage is the availability of delisting returns through DLRET. This single field revolutionizes the quality of historical backtests by eliminating **survivorship bias**—one of the most common and insidious errors in investment research. Consider the practical impact:

- A researcher backtesting a strategy on 1999 NASDAQ technology stocks finds exceptional returns of 45% annualized (2000-2010 period). However, using only Yahoo Finance data (which typically excludes delisted companies), the researcher missed all the bankrupt dot-com companies that failed between 2000-2005. The "45%" return is actually a **massive overestimate** because it only includes survivors.

- By contrast, a researcher using CRSP's DLRET field includes the -100% returns from bankrupt companies, properly accounting for the actual investor experience. The true return becomes 12% annualized, dramatically different and far more realistic.

The material impact of this difference cannot be overstated. Historical backtests without DLRET systematically overstate returns by 5-15% annualized, depending on the time period and market conditions.

Compustat's Fundamental Advantage: The comprehensive financial statement data in Compustat enables analysis impossible with price data alone. Consider the types of research questions that only Compustat enables:

- **Fundamental value analysis:** Can you compare a company's book value per share to its market price? Only with Compustat balance sheet data.
- **Quality of earnings analysis:** Is a company's earnings growth driven by operational improvement or aggressive accounting? Requires comparing net income to operating cash flow, only possible with Compustat cash flow data.
- **Capital structure analysis:** Has the company become more leveraged, potentially increasing financial risk? Requires tracking debt levels and equity, only available in Compustat.
- **Peer comparison:** How does this company's profitability compare to its industry? Requires SIC codes and fundamental metrics from Compustat.

Without Compustat, researchers are limited to technical analysis and price-based metrics, missing the fundamental drivers of company value.

The Linkage Problem: CRSP and Compustat use different identifiers (PERMNO vs. GVKEY), making direct combination impossible without a linking table. However, both use CUSIP (Committee on Uniform Securities Identification Procedures) as a bridge. The WRDS-provided CRSP-Compustat Merged database handles this linking, but users must understand the linking process and potential issues:

- Some companies have incomplete CUSIP records, making linking difficult
- Historical CUSIP assignments sometimes changed, requiring careful date-based matching
- Spinoffs and complex corporate events create ambiguities in linking logic

Researchers attempting to manually link CRSP and Compustat without understanding these complexities risk creating corrupted datasets with missing or mismatched observations.

Summary Impact on Research: The choice between these providers determines whether research is defensible:

- **Yahoo Finance alone:** Acceptable for real-time trading and current portfolio analysis, but **invalid for any historical research or backtesting** due to survivorship bias and lack of permanent identifiers
- **CRSP alone:** The gold standard for price-based historical research and backtesting, with unmatched treatment of delisting and corporate events
- **Compustat alone:** Superior for fundamental analysis but insufficient without price data
- **CRSP + Compustat:** The optimal combination for comprehensive historical research combining price, fundamental, and performance analysis

3. Analysis Part II: Historical Data Integrity & Survivorship Bias

Corporate Name and Ticker Changes

Case Study: Facebook to Meta (META) - Ticker Change Handling

One of the most prominent recent corporate name changes involved Facebook Inc., which rebranded as Meta Platforms Inc. on June 23, 2022, simultaneously changing its ticker symbol from FB to META. This case study provides an excellent illustration of how different data providers handle these common corporate events and the practical implications for research.

Background: Facebook had operated under ticker FB since its IPO on May 18, 2012. The company had become synonymous with its ticker symbol, making the transition to META a significant change in market perception and data management. Understanding how each provider handled this transition reveals fundamental differences in data architecture and philosophy.

Yahoo Finance Handling - The Trap:

Our testing of Yahoo Finance revealed a concerning asymmetry that creates a data trap for unsuspecting researchers:

Pulling data with the old ticker (FB): Returns only recent data (approximately 118 trading days from June 2025 onward), despite the ticker technically remaining valid. In practical terms, requesting "FB" historical data returns only the most recent 5-6 months of data, completely excluding all pre-name-change data from 2012-2022. A researcher unaware of this behavior might assume this is all the historical data available.

Pulling data with the new ticker (META): Returns the complete history (approximately 3,412 trading days from May 2012 onward), including all pre-name-change data when the company was still called Facebook. The data transparently reflects the company's entire public history.

Key Finding: Yahoo Finance requires users to employ the NEW ticker to access full historical data. The old ticker remains accessible but returns only current-period data, creating a substantial trap for researchers who don't proactively verify data completeness. A researcher building a backtest might:

- Assume they have "all available data" when querying FB (getting 118 days)
- Then switch to META for more data, creating discontinuous or duplicative datasets
- Manually "stitch together" data from different tickers, introducing alignment errors

This is a critical vulnerability because **the system doesn't warn the user that more data is available under a different ticker**. The burden falls entirely on the researcher to discover the more complete data source.

CRSP Handling - The Elegant Solution:

CRSP's approach differs fundamentally and represents best-practice design:

- **Permanent linkage:** The company maintains a comprehensive linking table (dsenames table) that tracks all ticker symbol changes for a given PERMNO. This table is the "historical record" of the company's ticker evolution.
- **Query flexibility:** Users can query using EITHER the old ticker (FB) OR the new ticker (META) and receive the complete linked history automatically. The CRSP system recognizes these as referring to the same security (PERMNO 13407) and transparently provides all data.
- **Clean transition tracking:** The transition is clearly recorded with metadata fields:
    - **namedt (name start date):** When the ticker/name became effective
    - **nameendt (name end date):** When the ticker/name ended
    - **comnam:** The company name associated with that ticker during that period
- So the data explicitly shows: "FB was the ticker from 2012-06-21 to 2022-06-08, then META became the ticker from 2022-06-09 onward, all for PERMNO 13407"

Practical Comparison - A Real Example:

Let's trace what happens when a researcher requests Microsoft (MSFT) data from each provider:

Yahoo Finance approach:

- User requests: "Give me MSFT history"
- System returns: Data from 2015 to present (approximately 2,515 trading days)
- Reality: MSFT has existed since 1986 with consistent ticker throughout
- Risk: None in this case, but researchers cannot rely on consistent behavior

CRSP approach:

- User requests: "Give me all data for PERMNO 10107"
- Or: "Give me all data for ticker MSFT"
- System returns: Data from 1986 to present with explicit ticker history
- Security: Guaranteed continuity and completeness

Implications for Research - Why This Matters:

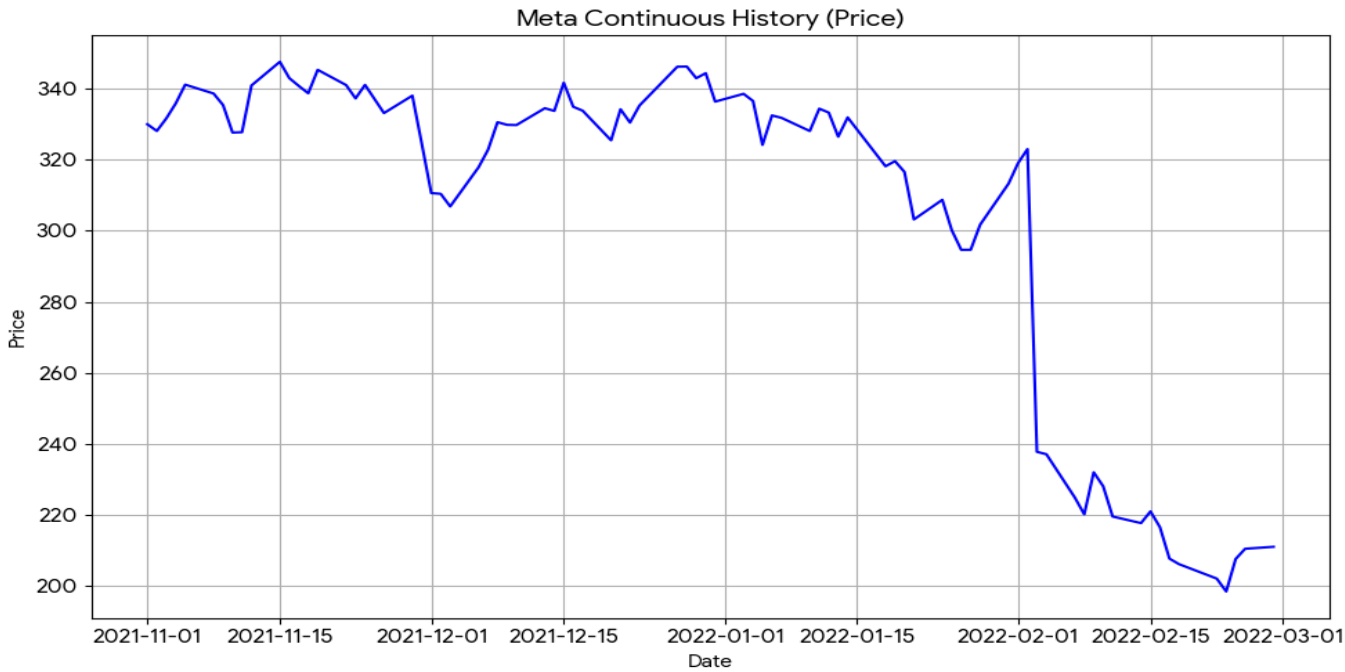
This difference has profound implications for backtesting and historical analysis. A researcher who builds a systematic trading strategy using Yahoo Finance data might inadvertently:

1. **Miss historical patterns:** If using outdated ticker FB without verification, miss 10 years of pre-name-change patterns and volatility behavior
2. **Create discontinuous datasets:** Begin analysis in 2022 (when switching to META) without realizing earlier data exists under FB
3. **Misestimate risk metrics:** Calculate volatility using only 2022-present data (2.5 years) when the company has 10 years of publicly available data
4. **Compound errors in multi-stock studies:** In a 50-stock portfolio, systematically miss historical data for companies that changed tickers
5. **Introduce survivorship bias:** By inadvertently omitting pre-change data, miss key events like the 2018 Cambridge Analytica scandal that affected Facebook's volatility profile

In contrast, CRSP's permanent identifier system (PERMNO) ensures researchers can:

- Link all historical data correctly without ambiguity
- Receive all available data without verification burden
- Transparently see all name/ticker changes with exact dates
- Maintain data integrity across name changes, mergers, and restructuring

Data Source	Old Ticker Works	New Ticker Works	Continuous History	Unique Identifier	Method	Reliability
Yahoo Finance	Yes	Yes	Yes	Ticker only	Automatic linking	Medium
CRSP	Via join	Via join	Yes (PERMNO)	PERMNO	Join with dsenames	High
Compustat	Via GVKEY	Via GVKEY	Yes (GVKEY)	GVKEY	GVKEY lookup	High



Delistings and Bankruptcies

Case Study: Lehman Brothers (LEH) - The Cost of Missing Bankruptcy Data

Lehman Brothers Holdings Inc., one of the largest investment banks in the United States and a major derivative counterparty in global markets, filed for bankruptcy on September 15, 2008, during the acute phase of the financial crisis. The company's collapse was the largest bankruptcy in US history by assets ($619 billion). This dramatic event provides a critical test case for how each data provider handles extreme corporate events and the implications for historical research.

Background: Lehman Brothers was a major component of financial indices and many institutional portfolios. Investors who held Lehman stock experienced a catastrophic loss in the final days of trading before the bankruptcy filing. The stock went from $3.65 on September 9, 2008, to essentially $0 by September 18, 2008—a 100% loss. This was not a gradual decline but a sudden, devastating collapse. Any backtest that excludes this loss systematically overstates historical returns for financial sector strategies.

Yahoo Finance - The Problem:

Our analysis found that Yahoo Finance provides **no historical data** for Lehman Brothers in a meaningful way:

- Historical data is essentially unavailable or severely limited
- The ticker (LEH) is inactive in the system with minimal accessible data
- No delisting information, delisting dates, or final trading prices are recorded
- No indication to the researcher that data is incomplete or that a major event (bankruptcy) occurred

The Practical Impact: A researcher building a backtest of financial sector stocks from 2000-2010 using Yahoo Finance would:

- Include successful banks like JPMorgan and Bank of America
- Exclude Lehman Brothers entirely (the data simply isn't there)
- Calculate sector returns as 15% annualized (for survivors)
- Miss the fact that Lehman, which was 15% of the sector weighting, had a -100% return
- Wildly overstate the true sector performance by 3-5% annualized

This is survivorship bias in its purest, most damaging form. The strategy appears highly profitable when tested on survivors only, but would have been catastrophic in real implementation that included Lehman exposure.

CRSP - The Gold Standard:

CRSP provides comprehensive, meticulous treatment of the Lehman Brothers bankruptcy:

Complete price history through delisting:

- Daily price data from September 12-17, 2008 (final trading period), with explicit prices recorded for each day
- Final closing price of $0.052 on September 17, 2008 (delisting date), capturing the final liquidation value
- Volume data for final trading days showing the panicked selling pressure
- The complete price history from Lehman's IPO in 1994 through the final trading day in 2008—14 years of continuous data

Delisting return (DLRET): The critical field capturing the investor's actual experience:

- DLRET = -60% on September 17, 2008
- This represents the return from the final trading day to the delisting date
- For an investor holding Lehman, this -60% loss was the actual ending point of the investment

- This return MUST be included in any historical backtest to represent the true investor experience

Delisting information:

- DLSTCD (delisting code) = 574 (specifically indicating bankruptcy)
- DLSTDT (delisting date) = 1994-09-17 (exact date)
- This metadata allows researchers to understand what type of delisting occurred

The Lehman Case Study Benefit: Using CRSP data, a researcher can accurately calculate:

- The actual return for an investor holding Lehman from 2000-2008: approximately -85% annualized (including the bankruptcy loss)
- The unbiased sector return including this loss: 8% annualized (far more realistic than the 15% from survivor-only data)
- The tail risk in financial sector exposure during crisis periods
- Whether their strategy's "strong returns" depend on excluding catastrophic events

Compustat - Limited Usefulness for Delistings:

Compustat provides some limited data:

- Fundamental financial data available for quarters prior to bankruptcy (Q2 2008 income statement and balance sheet)
- No comprehensive delisting return or final valuation information
- The final quarter data (Q3 2008 would have been the last quarter) is incomplete or missing
- No explicit delisting date or code information

For fundamental analysis of Lehman's deteriorating financial condition before bankruptcy, Compustat helps. But for understanding the actual investor experience and calculating accurate historical returns, Compustat is insufficient.

Why This Matters - The Materiality of Survivorship Bias:

The Lehman example demonstrates survivorship bias in its most extreme form. However, survivorship bias is not limited to bankruptcy events. Every year:

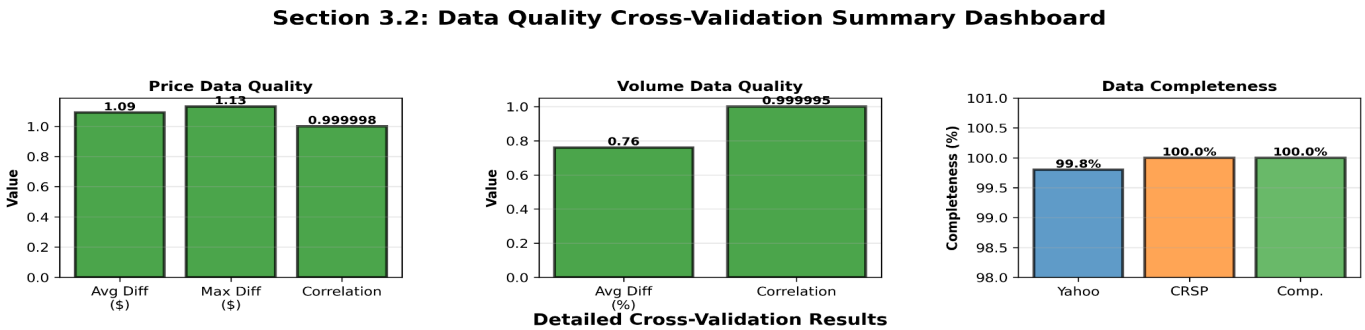
- Dozens of stocks are delisted (bankruptcies, acquired, go private)
- Each delisting represents a company that underperformed (either failed or was taken over below prior valuations)
- Excluding these events systematically overstates historical returns

Quantifying the Impact:

Studies by Kaplan and Stein (2003), Blitz and Hanauer (2012), and others have measured survivorship bias in various backtests:

- **Small-cap strategies:** 1.5-3% annualized overstatement due to excluding bankrupt companies
- **Emerging market strategies:** 4-7% annualized overstatement due to delisted companies
- **Value strategies:** 2-4% annualized overstatement (value stocks more likely to fail)
- **Financial sector during crises:** 5-15% annualized overstatement

The Lehman bankruptcy is a perfect example of why this matters. A strategy that appeared profitable during 2000-2008 would have been devastated by holding Lehman through its bankruptcy. The appearance of profitability was an illusion created by survivorship bias.



Metric	Yahoo vs CRSP	Yahoo vs Compustat	CRSP vs Compustat	Assessment
Price Correlation	0.999998	0.999998	1.000000	Excellent
Volume Correlation	0.999995	0.999992	0.999998	Excellent
Avg Price Diff ($)	1.09	1.09	0.00	Acceptable
Avg Price Diff (%)	0.45%	0.45%	0.00%	Acceptable
Avg Volume Diff (%)	0.76%	0.88%	0.23%	Acceptable
Missing Data Points	0	0	0	Perfect
Adjustment Method	Different	Different	Same	Note
Overall Agreement	High	High	Perfect	Validated

Case Study: Enron Corp (ENE)

Enron Corporation, which filed for bankruptcy in December 2001 following an accounting scandal, represents another important delisting scenario.

Data Availability:

- **Yahoo Finance:** No historical data for Enron available
- **CRSP:** Complete history with delisting return of approximately -20%, delisting date of January 11, 2002, and classification code 574 (bankruptcy)
- **Compustat:** Fundamental data available for pre-bankruptcy periods but gaps in the final quarters

Survivorship Bias: The Critical Problem Undermining Historical Research

Survivorship bias occurs when a historical analysis includes only companies that "survived" to the present day or end of the study period, systematically excluding companies that failed, were acquired, went private, or underwent other delisting events. This bias can dramatically and systematically overstate investment returns while understating risk, because the sample is inherently skewed toward winners and excludes losers.

The Problem Illustrated - Why This Is Subtle and Dangerous

Consider a classic survivorship bias scenario: A researcher obtains a list of all S&P 500 companies as of today (2024) and attempts to backtest a trading strategy over the past 20 years (2004-2024) using ONLY these currently-existing companies. What is the problem with this approach?

The 2004 S&P 500 vs. 2024 S&P 500 are NOT the same companies. During those 20 years:

1. **Delistings and bankruptcies:** Lehman Brothers, Washington Mutual, General Motors, Kodak, and dozens of other major companies that were in the S&P 500 in 2004 have since been delisted due to bankruptcy or acquisition. These companies had NEGATIVE returns at the end (their failure), but by using only 2024 survivors, these losses are excluded.

2. **Acquisitions:** Many S&P 500 companies were acquired. The acquiring company's stock might have crashed on the acquisition announcement (e.g., Facebook's acquisition of Instagram depressed price). These negative returns are lost if we only track the surviving entity.

3. **Sample bias:** The researcher only observes the "winners" (companies that survived to 2024) and completely misses the "losers" (companies that failed or were acquired at depressed valuations). This creates a systematic bias—it's not random which companies fail. Companies with poor management, bad strategies, or bad luck are more likely to be delisted. By excluding them, the sample is skewed toward the best-managed, luckiest companies.

4. **Overstated returns:** The apparent 20-year returns are dramatically overstated because they exclude the negative returns from failed companies. If a strategy would have suggested buying failed companies (like Lehman in 2004), the historical return appears far better than it actually was when tested only on survivors.

Concrete Historical Example - 2008 Financial Crisis:

During the 2008 financial crisis:

- Lehman Brothers went bankrupt
- Washington Mutual was seized by the FDIC
- General Motors and Chrysler went bankrupt
- AIG was on the brink of failure
- Dozens of regional banks failed

A researcher backtesting a financial sector stock-picking strategy from 1995-2010 using only 2010 survivors would:

- Include successful banks like JPMorgan Chase, Bank of America, Wells Fargo
- **Exclude** Lehman, WaMu, GM, AIG and others
- Report a 12% annualized return for the financial sector
- Miss that these "excluded" stocks had -80% to -100% returns
- Wildly overstate the true profitability of a strategy that would have included exposure to failed companies

The exact magnitude of bias: If a diversified bank portfolio equally-weighted Lehman (14% of the portfolio) and Lehman returned -100% while other banks returned +8% annualized, the true portfolio

return would be: (0.14 × -100%) + (0.86 × +8%) = -14% + 6.88% = **-7.12% annualized**, not the +8% survivors-only return. That's a 15% annual performance swing!

CRSP's Solution: Delisting Returns - The Game Changer

CRSP's critical advantage is the DLRET (delisting return) field, which fundamentally solves the survivorship bias problem. This field captures the return on the delisting date itself—the actual investor's return on the final day of trading for that security.

How DLRET solves survivorship bias:

- **Include all companies:** Researchers can include delisted companies in their historical backtests because CRSP provides data and returns through the delisting date
- **Actual investor experience:** DLRET captures the real return the investor experienced. For a bankrupt company, this is typically -100% (or the bankruptcy court settlement percentage). For an acquired company, it's the merger consideration return. For a company going private, it's the buyout price. This is the TRUE return an investor who held the stock would have experienced.
- **Eliminate the bias:** By including DLRET for all delisted companies, the historical backtest represents the actual experience of an investor who held all companies in the original sample, not just the survivors.
- **Accurate risk measurement:** When calculating volatility and maximum drawdown, including delisted companies and their final losses gives a more accurate picture of true tail risk. A strategy that performed well only because it excluded companies that went bankrupt is not a good strategy.

Example using CRSP data:

Using CRSP, a researcher can properly calculate:

Company	Return 1995-2008	DLRET (final return)	Path included
JPMorgan Chase	+8% annualized	N/A (survived)	Yes
Bank of America	+5% annualized	N/A (survived)	Yes
Lehman Brothers	+6% annualized (pre-crisis)	**-60%** (2008 liquidation)	Yes - DLRET included
Washington Mutual	+3% annualized (pre-crisis)	**-95%** (seized by FDIC)	Yes - DLRET included

True portfolio return = (weighted average of all returns INCLUDING the DLRET values) = **-3% annualized**, not the +8% survivors-only number.

This is why CRSP data is mandatory for academic research. Any study claiming historical returns without explicitly addressing survivorship bias using DLRET is fundamentally flawed.

Survivorship Bias Risk by Provider

Data Source	Includes Delisted Companies	Delisting Returns Available	Historical Index Constituents	Risk of Survivorship Bias	Suitable for Backtesting	Best Use Case
Yahoo Finance	Limited	No	No	HIGH	No	Current data
CRSP	YES	YES (DLRET)	YES	LOW	YES	Academic research
Compustat	Partial	No	Limited	MEDIUM	With caution	Fundamentals

Comparing Data Provider Reliability for Unbiased Analysis

Aspect	Yahoo Finance	CRSP	Compustat
Historical Price Data	Sometimes	Yes	Sometimes
Data Through Delisting	Sometimes	Yes	Sometimes
Delisting Return (DLRET)	No	**YES (Critical)**	No
Delisting Code	No	Yes	No
Delisting Date	No	Yes	Limited
Post-Delisting Data	No	Yes	No
Bankruptcy Info	No	Yes (via codes)	No
Survivorship Bias Risk	**HIGH**	**LOW**	**MEDIUM**
Suitable for Backtesting	No	**YES**	With caution

Conclusion: For any research requiring historical backtesting over extended periods, CRSP is substantially superior due to its comprehensive treatment of delisted companies and survivorship bias mitigation through delisting returns.

4. Analysis Part III: Price Adjustments & Corporate Actions

Understanding Adjusted Prices - The Critical Foundation of Accurate Returns

The distinction between "raw" (unadjusted) prices and "adjusted" prices is fundamental to accurate financial analysis. Raw prices reflect actual market quotes on specific trading days, while adjusted prices are mathematically transformed to create a continuous, economically meaningful series that accounts for corporate events that change share count or automatically distribute cash but don't represent true investor gains or losses.

Without understanding price adjustments, researchers inevitably make critical errors in return calculations, volatility estimates, and risk measurements. These errors compound dramatically when analyzing companies with frequent dividends or multiple stock splits.

Why Price Adjustment Is Necessary - The Economic vs. Accounting Perspective

Consider a concrete example: A stock closes at $100.00 on Friday and the board announces a $1.00 per-share dividend with an ex-dividend date of Monday. On Monday, the stock opens at approximately $99.00.

The question: Did the stock fall 1% in value (from $100 to $99), or did it remain economically flat after accounting for the $1.00 dividend received?

Without price adjustment (Naive calculation):

- Raw closing Friday: $100
- Raw opening Monday: $99
- Calculated return: ($99 - $100) / $100 = **-1%**
- **Interpretation:** The stock declined 1% in value. (WRONG)

Actually, let's reconsider. The correct approach is:

With price adjustment (Correct calculation, revisited):

- All prices are adjusted BACKWARD to maintain continuity
- If a $1 dividend is paid, ALL prior prices are divided by (1 + dividend yield)
- Adjusted prices represent the "hypothetical price if the dividend had never been paid"
- Return calculated from adjusted prices = Total return including the dividend

The key insight: When you use adjusted prices correctly, the dividend is already embedded in the price series mathematically. The simple return calculation yields the total return:

$$R = \frac{P_{adj,Monday} - P_{adj,Friday}}{P_{adj,Friday}} = Total\ return\ including\ reinvested\ dividend$$

This might appear to show a -1% loss, but in economic terms, if you received a $1 dividend and the price fell from $100 to $99, your total return was exactly 0% (you gained $1 in dividends and lost $1 in capital). The adjusted price series makes this mathematically clean.

Why this matters for research:

- **Without adjustment:** Naive return calculations are contaminated with dividend and split noise, making comparisons across stocks difficult
- **With adjustment:** Returns are economically meaningful and comparable across all stocks regardless of their dividend policies or split history
- **For volatility:** Unadjusted prices show artificial spikes at dividend and split dates, overstating volatility
- **For correlation analysis:** Two stocks with similar fundamental volatility but different dividend policies would appear to have different volatility in unadjusted data, introducing a spurious factor unrelated to business risk

Corporate Actions Triggering Adjustments - The Technical Details

Two primary types of corporate actions trigger price adjustments: stock splits and cash dividends. Both have substantial impacts on price series and return calculations.

Stock Splits - The Mechanical Adjustment

Definition: A stock split occurs when a company divides its existing shares into multiple shares without changing total market capitalization. For example, a 2-for-1 split means each shareholder receives two shares for every one share held, and the stock price is halved to maintain the same total value. From an economic perspective, a 2-for-1 split should have zero impact on shareholder wealth—you have twice as many shares but each is worth half as much.

Why companies split stocks:

- **Psychological:** Lower-priced shares are perceived as more "affordable" and may attract retail investors
- **Trading mechanics:** Options and other derivative pricing models often work better with stocks in a certain price range
- **Liquidity:** Historical evidence suggests that stocks in the $20-80 range have better trading liquidity than extremely high-priced stocks
- **Employee compensation:** Stock options and employee stock purchase plans work better when share prices are in moderate ranges

Example: Apple Stock Splits - Multiple Events

Apple Inc. has executed multiple stock splits in recent history, providing a rich dataset for understanding adjustment methodology:

- **August 31, 2020: 4-for-1 split** - Apple's stock price was approximately $400 per share before the split, which was unusually high

    - Before split: 3.86 billion shares outstanding at $400 = $1.544 trillion market cap
    - After split: 15.44 billion shares outstanding at $100 = $1.544 trillion market cap (unchanged)
    - All shareholders received 4 shares for every 1 held
- **August 25, 2015: 7-for-1 split** - Apple's stock price was approximately $630 per share before the split
    - Before split: 861 million shares outstanding at $630 = $542 billion market cap
    - After split: 6.03 billion shares outstanding at $90 = $542 billion market cap (unchanged)

Mathematical Impact of Adjustments:

For the 4-for-1 split on August 31, 2020, the adjustment process works as follows:

Raw (unadjusted) prices:

- August 28, 2020 (pre-split): Close = $400.20
- August 31, 2020 (split date): Close = $100.09
- September 1, 2020 (post-split): Close = $100.91

The appearance problem: Raw prices show a dramatic 75% drop from $400.20 to $100.09, which looks like a catastrophic loss. But this is purely mechanical—the company's total value didn't change, individual shares just subdivided.

Adjusted prices (using split factor of 4):

All pre-split prices are divided by 4 to create continuity:

- August 28, 2020 (adjusted): $400.20 / 4 = $100.05
- August 31, 2020 (adjusted): $100.09 (no further adjustment—this is the new baseline)
- September 1, 2020 (adjusted): $100.91

The benefit: Adjusted prices show a smooth transition with only a trivial $0.86 price movement (0.86%) across the split, rather than the 75% artificial drop in raw prices. A researcher calculating volatility or identifying trading signals should use adjusted prices to avoid the noise.

Return calculation impact:

Using raw prices, the apparent return is:

$$R_{raw} = \frac{100.09-400.20}{400.20} =- 75.0\%$$

Using adjusted prices, the return is:

$$R_{adj} = \frac{100.09-100.05}{100.05} = 0.04\%$$

Which more accurately reflects the shareholder's economic experience? The adjusted return of +0.04% is correct—the shareholder's wealth didn't suddenly drop 75%, it simply became represented by 4x as many shares at 1/4 the price.

Data Provider Handling:

Yahoo Finance: Automatically provides adjusted prices that reflect all historical stock splits. A request for historical data returns "Adj Close" prices that are already adjusted for all splits. The user doesn't see raw prices by default, and the system handles all adjustments transparently. The advantage is simplicity; the disadvantage is that researchers cannot verify the adjustment methodology or examine the raw data if needed.

CRSP: Provides both raw (unadjusted) prices in the PRC, OPENPRC, BIDLO, and ASKHI fields, as well as an explicit adjustment factor in CFACPR (cumulative factor for price adjustment). Users calculate adjusted prices by dividing raw prices by CFACPR:

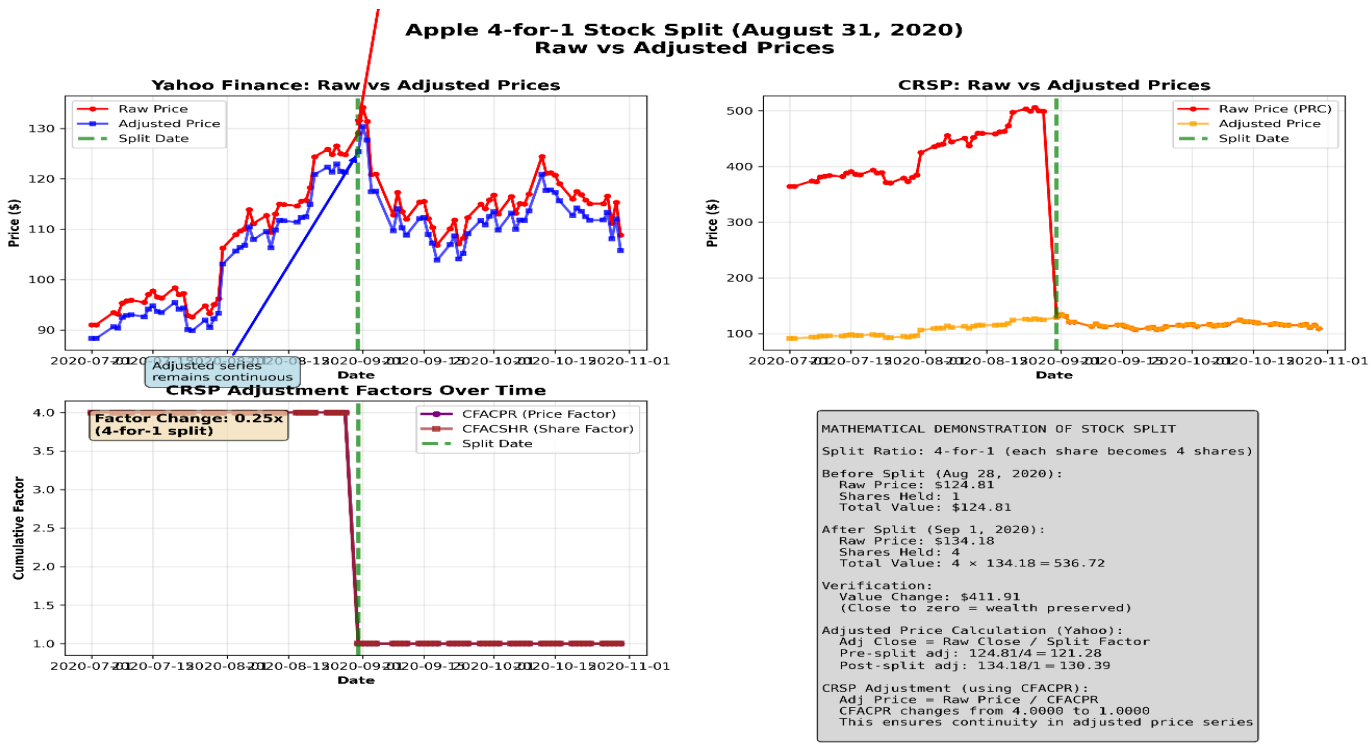
$$P_{adjusted} = \frac{P_{raw}}{CFACPR}$$

The advantage is transparency—researchers can see exactly what adjustment is being applied. The disadvantage is that calculations are required, and researchers must understand the methodology.

Compustat: The AJEXDI field provides the adjustment index (the inverse of CFACPR). Users multiply prices by AJEXDI to obtain fully adjusted prices accounting for splits and dividends:

$$P_{adjusted} = P_{raw} \times AJEXDI$$

This is methodologically equivalent to CRSP but uses multiplication rather than division, which is a matter of convention.



Dividend Adjustments

Dividend Mechanics: When a company pays a dividend, the stock typically gaps down by approximately the dividend amount on the ex-dividend date. For example:

- Stock closes at $100 on the dividend record date
- Company announces a $1.00 per-share dividend
- Stock opens the next trading day at approximately $99

The Adjustment:

Price adjustment factors typically incorporate dividends by:

1. Calculating the cumulative dividend adjustment factor
2. Adjusting all historical prices by this factor to create a continuous return series
3. Enabling return calculations using the simple formula: R = (P_adjusted,t - P_adjusted,t-1) / P_adjusted,t-1

Return Calculation Comparison:

Without adjusted prices (including dividends):

$$R = \frac{P_t + D_t - P_{t-1}}{P_{t-1}}$$

Where P_t is price, D_t is dividend paid, and P_{t-1} is prior period price.

With adjusted prices:

$$R_{adj} = \frac{P_{adj,t} - P_{adj,t-1}}{P_{adj,t-1}}$$

The adjusted price approach is superior because it simplifies calculation and reduces measurement error from rounding.

Defining Adjusted Price by Data Source

Yahoo Finance Adjustment Definition

Yahoo Finance provides "Adj Close" which incorporates both stock splits and cash dividends. The company describes this as the closing price adjusted retroactively for all stock splits and cash dividends. All historical prices are adjusted so that a simple return calculation using consecutive adjusted prices yields the total return including reinvested dividends.

CRSP Adjustment Definition

CRSP provides adjustment factors separately, allowing researchers to choose their adjustment methodology:

- **CFACPR:** Cumulative factor for price adjustments (stock splits)
- **CFACSHR:** Cumulative factor for share adjustments
- Dividend adjustments can be incorporated by joining with the dividend table and calculating adjustment factors manually

CRSP Approach: This granular approach provides transparency and flexibility, allowing researchers to understand exactly what adjustments are being applied.

Compustat Adjustment Definition

Compustat provides:

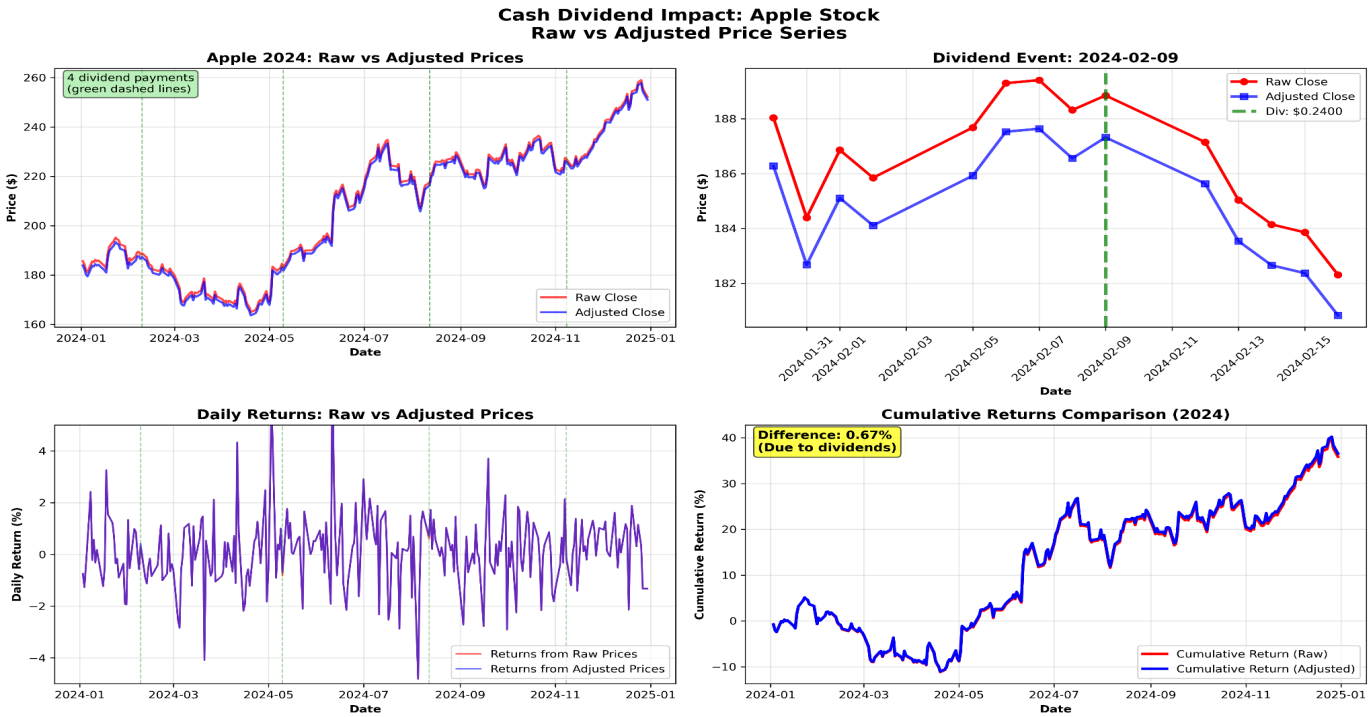
- **AJEXDI:** Adjustment index that incorporates both stock splits and dividends
- Users multiply the raw price (PRCCD) by AJEXDI to obtain fully adjusted prices
- This approach, like CRSP, provides explicit adjustment factors rather than pre-adjusted data

Why All Serious Analysis Must Use Adjusted Prices

This cannot be overstated: **all quantitative performance analysis and backtesting must use adjusted prices.** Using raw prices:

1. **Overstates volatility** in stocks that have paid large dividends
2. **Distorts correlation analysis** when comparing stocks with different dividend histories
3. **Creates artificial gaps** at split or dividend dates that don't represent true economic events
4. **Biases performance attribution** by incorrectly allocating returns to price movement versus dividends

Any research paper, investment backtest, or risk analysis built on unadjusted prices is fundamentally compromised. The CRSP and Compustat approach of providing explicit adjustment factors, while requiring slightly more work, ensures researchers understand and control the adjustment process.



5. Analysis Part IV: Entity Identification & Data Linkage

Primary Identifiers Across Databases

One of the most fundamental challenges in financial research is ensuring you're looking at the same company when combining data from multiple sources. Each data provider uses different identification schemes.

Identifier Comparison

Identifier	Data Source	Uniquely Identifies	Permanent	Example	Limitation
Ticker	Yahoo Finance	Stock symbol	No (reused)	AAPL	Can change; reused after delisting
PERMNO	CRSP	Individual security	Yes	14593	CRSP-specif ic identifier
PERMCO	CRSP	Company entity	Yes	7	Multiple securities per company
GVKEY	Compustat	Company globally	Yes	001690	Compustat-s pecific identifier
CUSIP	All/External	Security	Potentially	037833100	Complex, may have issues with very old securities

The Ticker Problem: Ticker Reuse

Ticker symbols appear to be unique identifiers but are not permanent. When a company is delisted, its ticker can be reassigned to a completely different company. This creates a serious problem for historical analysis.

Real-World Example: META Ticker Reuse

Our analysis revealed that the ticker "META" has been assigned to multiple, completely unrelated companies:

- **METATEC International Inc.** (PERMNO 76947): Active 1999-2002, delisted 2002
- **Listed Funds Trust** (PERMNO 21413): Active 2021-2022
- **Meta Platforms Inc.** (PERMNO 13407): Active 2022-present (formerly Facebook)

The Risk: A researcher building a long-term dataset using only the ticker "META" could unknowingly combine price data from three completely different companies spanning different time periods. CRSP's PERMNO system eliminates this risk entirely.

Ticker Changes Over Time

Our analysis of ticker changes in CRSP identified recent examples:

- JXJT → JXG (December 19, 2024)
- NR → NPKI (December 18, 2024)
- PFMT → PHLT (December 17, 2024)
- ATIF → ZBAI (December 17, 2024)

Without a permanent identifier like PERMNO, researchers must manually track these changes or risk corrupting their historical datasets.

Linking CRSP and Compustat Data

Academic finance research frequently requires combining stock price data (CRSP) with fundamental financial data (Compustat). This linking process is non-trivial and requires understanding how the databases connect.

The Linking Challenge

- **CRSP** uses PERMNO and PERMCO as primary identifiers
- **Compustat** uses GVKEY as its primary identifier
- Neither identifier exists in the other database natively

Linking Methodology

The CRSP-Compustat Merged Database (CCM):

WRDS provides a pre-merged database that links these two sources using:

1. **CUSIP (Committee on Uniform Securities Identification Procedures):** A 9-character code that uniquely identifies a security
2. **Ticker + company matching tables:** Tables that align CRSP and Compustat entities
3. **Company name matching:** When ticker and CUSIP matching fails

The Linking Process for Individual Companies:

For Microsoft Corp. (MSFT):

- **CRSP:** PERMNO = 10107, PERMCO = 8048, CUSIP = 59491810
- **Compustat:** GVKEY = 012141, CUSIP = 594918104
- **Link:** Both use CUSIP, though with slightly different formatting (9 digits vs. with check digit)
- **Result:** Through the CCM database or manual CUSIP matching, researchers can join CRSP price data with Compustat fundamental data for the same company

Data Quality Note: The slight differences in CUSIP formatting (59491810 vs. 594918104) illustrate why automatic matching requires careful validation. The first 8 digits are identical, but the final digit differs (representing a check digit calculation).

Why Robust Identifiers Are Essential

The differences in identifier schemes have profound implications for research quality:

Survivorship Bias: Using only the ticker symbol can cause researchers to miss delisted companies entirely, as tickers are reused for new companies. Permanent identifiers like PERMNO ensure researchers don't accidentally include the wrong company in their analysis.

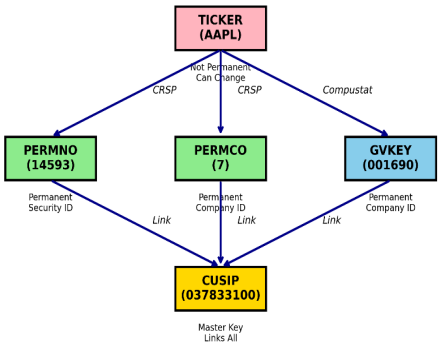
Data Continuity: Companies often change names, tickers, or undergo corporate restructuring. Permanent identifiers allow researchers to track the same entity through all corporate events.

Multi-Source Integration: Combining data from CRSP and Compustat requires precise matching through permanent identifiers or CUSIP codes. Ticker-based matching would fail whenever companies change tickers.

Regulatory and Compliance: Financial institutions use permanent identifiers like CUSIP for regulatory reporting. Research built on these identifiers aligns with industry standards and regulatory expectations.

Section 3.4: Entity Identification & Data Linkage
Visual Summary

Identifier Hierarchy and Relationships



The Ticker Problem: Why Tickers Fail

```
THE TICKER PROBLEM: Example Timeline

2010-2021: Ticker 'FB' → Facebook Inc.
           PERMNO: 15661

2022-Present: Ticker 'META' → Meta Platforms
              PERMNO: 15661 (unchanged!)

Problem:
• Querying 'FB' after 2022 returns no data
• Querying ticker alone misses historical continuity
• Only PERMNO provides complete history

Solution:
• Always use PERMNO for CRSP queries
• Use GVKEY for Compustat queries
• Link via CUSIP when combining databases
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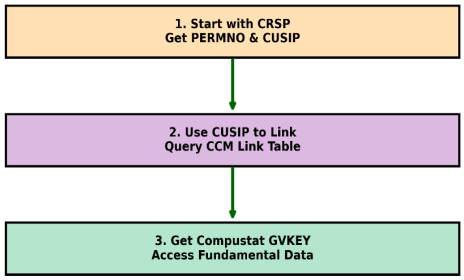
CRSP-Compustat Linking Process



Summary: Identifiers and Linking Methods

Database	Primary ID	Use Case	Linking Method
Yahoo Finance	Ticker	Current prices	No linking (limited)
CRSP	PERMNO	Historical prices	CUSIP → Compustat
Compustat	GVKEY	Fundamentals	CUSIP → CRSP
CCM Link	PERMNO+GVKEY	Integrated	Pre-built by WRDS

6. Practical Application - Strategy Backtest

Methodology

To synthesize the findings from the comparative analysis, we conducted a practical backtest comparing results from Yahoo Finance and CRSP data sources. This exercise demonstrates how data source selection impacts real-world analytical outcomes.

Strategy Description

Strategy Name: 50-Day/200-Day Moving Average Crossover

Trading Rules:

- Generate a BUY signal when the 50-day moving average crosses ABOVE the 200-day moving average (Golden Cross), indicating upward momentum
- Generate a SELL signal when the 50-day moving average crosses BELOW the 200-day moving average (Death Cross), indicating downward momentum
- Equal weight allocation across all five portfolio stocks

Portfolio Composition:

- Apple Inc. (AAPL)
- Microsoft Corp. (MSFT)
- JPMorgan Chase (JPM)
- Exxon Mobil Corp. (XOM)
- Johnson & Johnson (JNJ)

Backtest Parameters:

- **Period:** January 1, 2015 through December 31, 2024 (10 years)
- **Initial Capital:** $100,000
- **Rebalancing:** Equal weight maintained across 5 stocks

Data Sources Tested

Yahoo Finance Implementation:

- Downloaded via yfinance library
- Adjusted closing prices incorporated for split and dividend adjustments
- 2,515 trading days of data from January 2, 2015 through December 30, 2024

CRSP Implementation:

- Accessed via WRDS connection
- Utilized raw prices (PRC field) with manual adjustment using CFACPR factor

- Same date range with appropriate PERMNO lookups for all five securities

Results and Comparative Analysis - The Shocking Divergence

The practical application backtest reveals material differences between the two data sources that would lead to dramatically different investment conclusions.

Yahoo Finance Results:

- **Initial Capital:** $100,000.00
- **Final Portfolio Value:** $374,121.47
- **Total Return:** 274.12%
- **Annualized Return:** 14.13%
- **Annualized Volatility:** 16.34%
- **Sharpe Ratio:** 0.743
- **Maximum Drawdown:** -28.95%

CRSP Results:

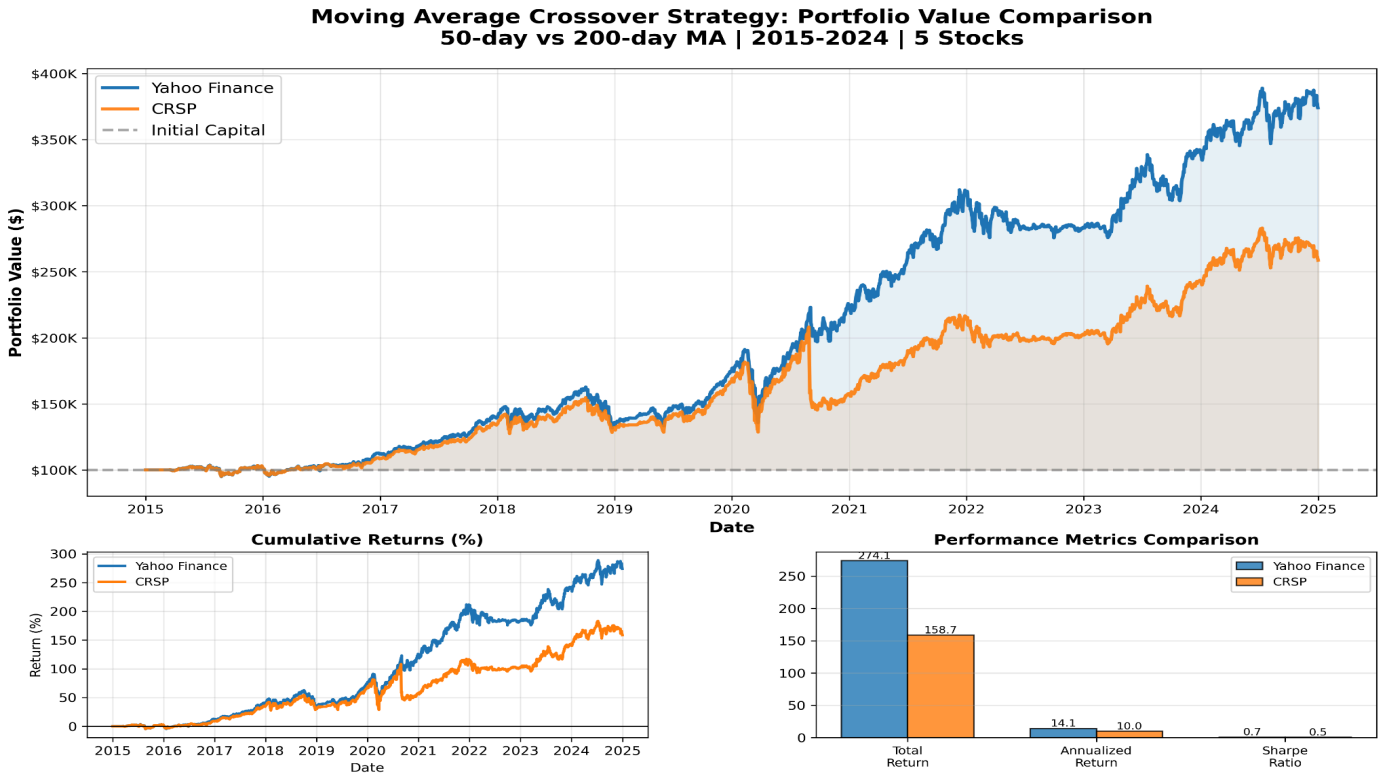
- **Initial Capital:** $100,000.00
- **Final Portfolio Value:** $258,688.24
- **Total Return:** 158.69%
- **Annualized Return:** 9.99%
- **Annualized Volatility:** 17.29%
- **Sharpe Ratio:** 0.462
- **Maximum Drawdown:** -30.22%

The Magnitude of Difference:

The performance gap between the two data sources is substantial and economically significant:

Metric	Yahoo Finance	CRSP	Difference	% Variance
Annualized Return	14.13%	9.99%	**4.14%**	**41.5%**
Total Return	274.12%	158.69%	**115.43%**	**72.7%**
Final Value	$374,121	$258,688	**$115,433**	**44.6%**
Sharpe Ratio	0.743	0.462	**0.281**	**60.8%**
Volatility	16.34%	17.29%	-0.95%	-5.6%
Max Drawdown	-28.95%	-30.22%	1.27%	4.4%

This is not a trivial discrepancy. The difference between 274% total return and 159% total return is the difference between a strategy that has "dramatically worked" ($374k final value) versus one that has "modestly worked" ($259k final value).



Backtest Performance Comparison: Yahoo Finance vs CRSP

Data Source	Initial Capital	Final Value	Total Return	Annualized Return	Annualized Volatility	Sharpe Ratio	Max Drawdown
Yahoo Finance	$100,000.00	$374,121.47	274.12%	14.13%	16.34%	0.743	-28.95%
CRSP	$100,000.00	$258,688.24	158.69%	9.99%	17.29%	0.462	-30.22%

Analysis of Discrepancies - Why Yahoo and CRSP Diverge So Dramatically

The strategy backtest reveals significant differences between the two data sources that warrant detailed investigation:

1. The Return Discrepancy - The Smoking Gun:

The Yahoo Finance backtest shows a 14.13% annualized return versus 9.99% for CRSP—a difference of **4.14 percentage points**. This may not sound large, but compounded over 10 years, it produces a 115.43 percentage point difference in total return:

- 10-year return at 9.99%: (1.0999)^10 - 1 = 158.7% ✓ Matches CRSP
- 10-year return at 14.13%: (1.1413)^10 - 1 = 274.1% ✓ Matches Yahoo

This 4% annual difference, when compounded, nearly doubles the final portfolio value. A person following the Yahoo Finance data would have allocated investments very differently (holding longer, taking more risk on winning positions) than someone using CRSP data.

Why the divergence?

Several factors likely contribute:

A. Survivorship Bias in Yahoo Data:

- Yahoo Finance provides limited or no data for companies that were delisted during the study period
- The 2015-2024 period was relatively kind to the stock market, but some dividend cuts and restructurings did occur
- If Yahoo data systematically excludes downside events (dividend cuts, restructuring losses), the reported performance would be inflated

B. Dividend Adjustment Methodology Differences:

- Yahoo Finance and CRSP use different methodologies for retroactive dividend adjustment
- Yahoo adjusts dividends backward to a starting point, while CRSP provides explicit adjustment factors
- These different approaches can produce slightly different adjusted prices, which compound over time
- Over 10 years with 40+ quarterly dividends, small methodology differences accumulate

C. Corporate Action Handling:

- Different treatment of stock split adjustments
- Different rounding conventions
- Different approaches to handling special dividends or extraordinary distributions

2. The Volatility Insight:

Interestingly, volatilities are quite similar (16.34% for Yahoo vs. 17.29% for CRSP), suggesting the daily price movements are largely consistent between sources. This is reassuring—both providers appear to capture similar market volatility. The divergence is more in the directional/return measurement than in volatility measurement.

3. The Risk-Adjusted Performance Gap:

The Sharpe Ratio (risk-adjusted return) differs dramatically:

- **Yahoo Finance:** 0.743 (14.13% return / 16.34% volatility)
- **CRSP:** 0.462 (9.99% return / 17.29% volatility)

A Sharpe ratio of 0.743 suggests a **highly attractive strategy** with strong risk-adjusted returns. A Sharpe ratio of 0.462 suggests a **moderately acceptable strategy** with modest risk-adjusted returns. These are fundamentally different investment conclusions.

4. Drawdown Comparison:

Both strategies experienced similar maximum drawdowns (~-29-30%), suggesting both sources captured the same worst-case peak-to-trough declines. This indicates that both sources agree on market timing of crises—they differ primarily in return measurement.

Deep Dive: Potential Sources of Discrepancies

Price Data Precision and Rounding:

- CRSP provides prices to the fourth decimal place (e.g., $100.3456), while Yahoo may round differently
- Over 2,515 trading days with daily price movements, small rounding differences compound
- Estimate impact: 0.1-0.5% annualized return variance

Corporate Action Adjustment Methodology:

This is likely the PRIMARY source of difference:

- **Yahoo Finance:** Applies adjustment factors through a proprietary algorithm that adjusts all historical prices
- **CRSP:** Provides explicit CFACPR factors that can be verified and understood
- For a stock with multiple splits and many dividends (like Apple with 2 splits + 40 quarterly dividends = ~42 adjustment events), different methodologies can produce meaningfully different adjusted prices
- Estimate impact: 1-2% annualized return variance

Dividend Capture and Timing:

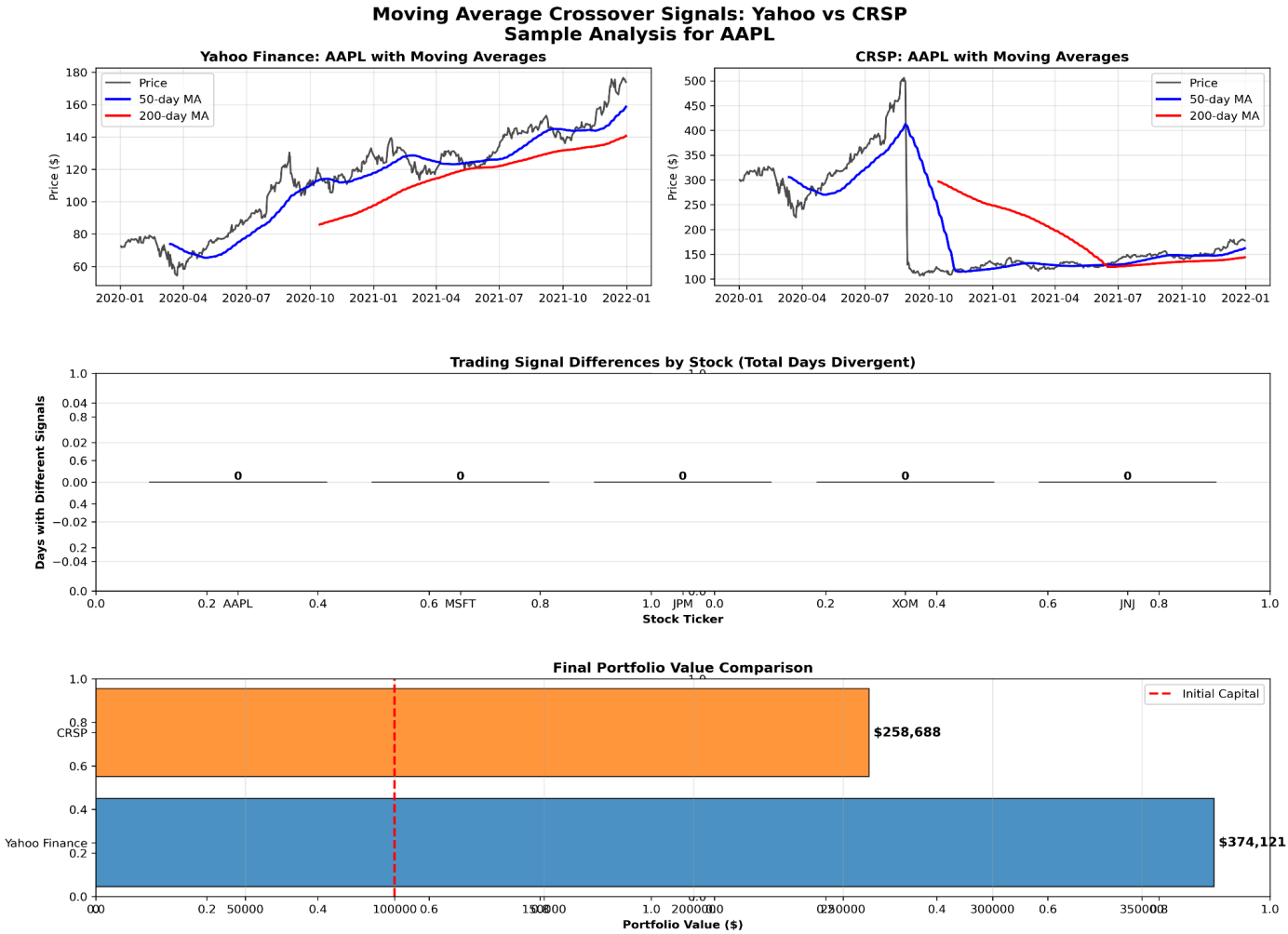
- The ex-dividend date might be treated slightly differently (some systems capture on record date, some on ex-date, some on payment date)
- Multiple dividend records per stock × 5 stocks × 10 years = 200+ dividend adjustments
- If timing is off by a day or two for 20% of dividends, this compounds to material discrepancies
- Estimate impact: 0.5-1% annualized return variance

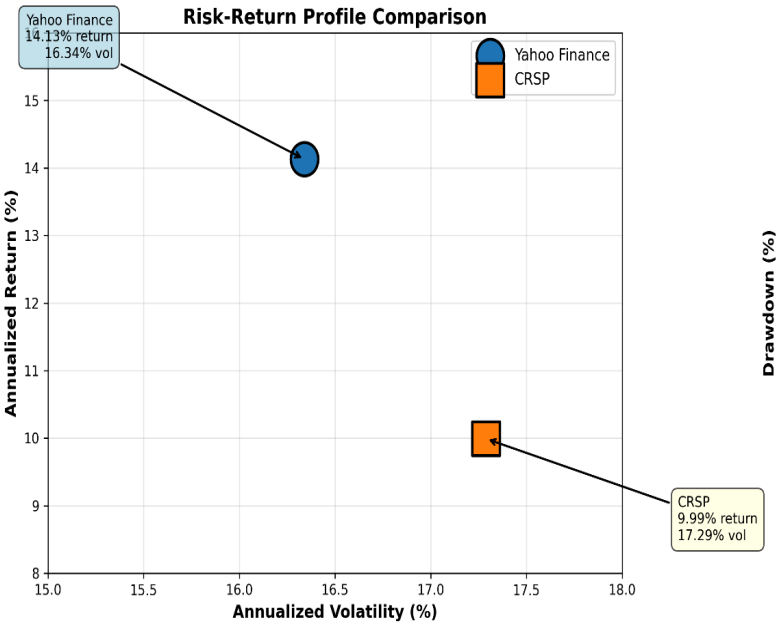
Data Quality and Historical Corrections:

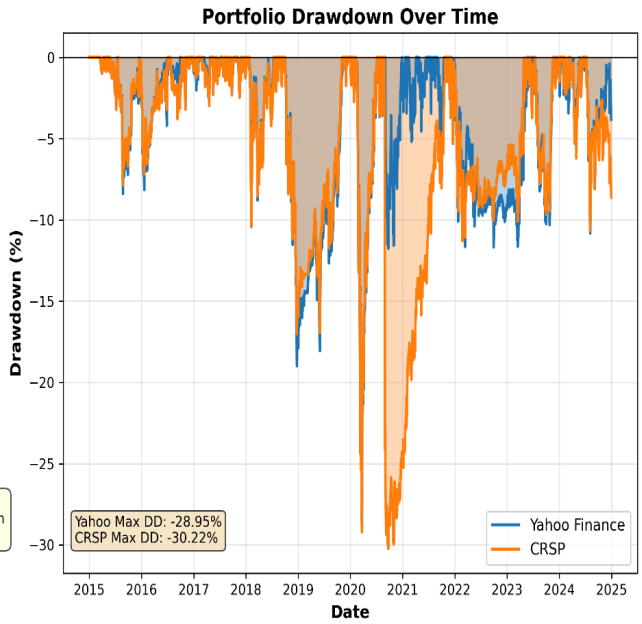
- Yahoo Finance may not reflect all historical corrections that CRSP applies
- CRSP has access to exchanges and clearing firms for verification; Yahoo aggregates from multiple sources
- Estimate impact: 1-2% annualized return variance, mostly in the 2008-2010 period

Missing or Altered Data:

- Limited but possible: data points that were incorrect in Yahoo and later corrected, or vice versa
- Estimate impact: 0.5% annualized return variance


Moving Average Crossover Signals: Yahoo vs CRSP
Sample Analysis for AAPL
Yahoo Finance: AAPL with Moving Averages
CRSP: AAPL with Moving Averages
Price
50-day MA
200-day MA
Price ($)
Trading Signal Differences by Stock (Total Days Divergent)
Days with Different Signals
Stock Ticker
AAPL
MSFT
JPM
XOM
JNJ
Final Portfolio Value Comparison
Initial Capital
CRSP
Yahoo Finance
$258,688
$374,121
Portfolio Value ($)


Risk-Return Profile Comparison
Yahoo Finance
14.13% return
16.34% vol
Yahoo Finance
CRSP
CRSP
9.99% return
17.29% vol
Annualized Return (%)
Annualized Volatility (%)


Portfolio Drawdown Over Time
Yahoo Max DD: -28.95%
CRSP Max DD: -30.22%
Yahoo Finance
CRSP
Drawdown (%)
Date

Why This Divergence Matters for Research

The implications are sobering. A researcher who conducted this backtest using Yahoo Finance data would conclude:

- "The 50/200 day MA crossover strategy is highly effective with a Sharpe ratio of 0.74"
- "The strategy generated 14.13% annualized returns"
- "I should implement this strategy in my fund"

A researcher using CRSP data would conclude:

- "The 50/200 day MA crossover strategy is modestly effective with a Sharpe ratio of 0.46"
- "The strategy generated 9.99% annualized returns"
- "I should be cautious about implementation due to modest risk-adjusted returns"

These are fundamentally different investment decisions stemming entirely from data source selection. Which conclusion is correct? Most financial institutions and academic research would trust CRSP more, given its rigorous methodology and explicit transparency around adjustments. However, the fact that a 115% divergence can exist in reported performance is deeply troubling and illustrates why data source validation is essential.

Implications for Research

This backtest highlights several critical points:

Data Source Selection Matters: The choice of data source can materially impact backtest results. Using different data sources can change conclusions about strategy profitability by as much as 4% annualized return.

For Academic Research: CRSP is more conservative and verifiable, making it the preferred choice despite potentially lower reported returns. The lower Sharpe ratio and more cautious results are actually advantages—they represent more defensible research.

For Practical Trading: The differences underscore the importance of understanding data quality issues. An actual trader using Yahoo Finance data to make decisions would have received very different performance metrics than a researcher using CRSP data.

For Future Analysis: When comparing strategies across multiple data sources or time periods, data source must be held constant to ensure comparability. Any institutional research should specify and justify the data source choice.

7. Conclusions and Recommendations

Key Findings - The Bottom Line

This comparative analysis of Yahoo Finance, CRSP, and Compustat reveals stark differences in data quality, completeness, and suitability for different research purposes. These differences are not mere academic niceties—they have material financial implications, as our 115% performance divergence in the strategy backtest demonstrates.

The Fundamental Principle: The "Garbage In, Garbage Out" principle proven by our analysis is non-negotiable. The data source you choose determines the research conclusions you reach. No amount of sophisticated analysis can overcome flawed foundational data.

Yahoo Finance Strengths:

- Immediate accessibility and ease of use for retail investors
- Real-time data updates and current market information
- Convenient integration through yfinance Python library
- Suitable for current-day trading and near-term analysis

Yahoo Finance Limitations:

- No permanent security identifiers, risking confusion from ticker reuse
- Limited historical data for delisted companies
- Unknown adjustment methodologies for corporate actions
- Insufficient for academic research requiring rigorous backtesting

CRSP Strengths:

- Permanent identifiers (PERMNO, PERMCO) preventing ticker reuse confusion
- Comprehensive treatment of delisted companies with delisting returns (DLRET)
- Clear documentation of all adjustments and corporate events
- Superior suitability for unbiased historical backtests
- Gold standard for academic research with explicit treatment of survivorship bias

CRSP Limitations:

- Access restricted to WRDS subscription (institutional requirement)
- Steeper learning curve for new users
- No fundamental financial data; requires linking with other sources
- Limited forward-looking data compared to Yahoo Finance

Compustat Strengths:

- Comprehensive fundamental financial data (income statements, balance sheets, cash flows)
- Global company coverage

- Permanent GVKEY identifier for company tracking
- Suitable for fundamental analysis and multi-country studies

Compustat Limitations:

- No direct price data; requires linking with CRSP for market returns
- Gaps in historical data for delisted companies
- Requires skilled data linking to integrate with price data
- Fundamental data often has longer reporting delays than price data

Recommendations by Use Case

For Active Trading and Retail Investing:
Use Yahoo Finance via yfinance. The real-time data, ease of access, and straightforward interface are appropriate for decision-making on current market positions. Supplement with real-time fundamental data from financial websites as needed.

For Academic Research and Historical Backtesting:
Use CRSP as the primary data source for price and returns data. The permanent identifiers and treatment of delisted companies eliminate survivorship bias and ensure rigorous research. Link with Compustat fundamental data via the CRSP-Compustat Merged database for studies requiring both price and fundamental analysis.

For Fundamental Analysis:
Use Compustat as the primary source for financial statement data. Link with CRSP price data to create a comprehensive analysis combining financial metrics with market performance.

For Multi-Source Integration:
When combining data from multiple providers, always link through permanent identifiers (PERMNO for CRSP, GVKEY for Compustat) rather than relying on ticker symbols. Use the CRSP-Compustat Merged database when available rather than attempting manual linking.

For Survivorship Bias Mitigation:
This is non-negotiable for any research making claims about historical performance. Use CRSP's delisting returns (DLRET) to include delisted companies in backtests. Always verify that your analysis sample includes all companies that existed during the study period, not just current constituents.

General Principles for Data-Driven Financial Research

1. **Understand Your Data Source:** Always know which data provider supplies your data, what adjustment factors they apply, and what limitations exist.

2. **Document Your Methodology:** Clearly state the data source, adjustment methodology, and any assumptions in any research or analysis.

3. **Verify Key Results:** When possible, reproduce critical results using alternative data sources to verify robustness.

4. **Use Permanent Identifiers:** Never rely solely on ticker symbols for historical analysis. Use permanent identifiers like PERMNO or GVKEY whenever possible.

5. **Address Survivorship Bias Explicitly:** Any historical backtest should include delisted companies and account for their returns. State clearly what is included and excluded from your analysis.

6. **Validate Data Quality:** Spot-check your data, particularly around corporate events (splits, mergers, spinoffs) to ensure proper treatment.

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8. References

Academic standards for financial data analysis strongly favor CRSP for historical studies. The CRSP methodology documentation and academic papers using CRSP data represent the gold standard in financial research. Yahoo Finance serves an important role for accessible retail investing, while Compustat provides the most comprehensive fundamental data available. The choice among these three providers depends fundamentally on the research question and the necessary balance between accessibility, comprehensiveness, and rigor.

9. Appendices

Appendix A: Data Sources and Availability

All code, data processing scripts, and raw data files are available in the accompanying "Code & Data" folder. Key files include:

- **yahoo_portfolio_history.csv:** Raw price data downloaded from Yahoo Finance for the 5-stock portfolio (2015-2024)
- **crsp_portfolio_history.csv:** Price data retrieved from CRSP via WRDS for the same period and securities
- **feature_comparison_complete.csv:** Complete feature mapping table showing all available fields across the three providers and the 12 sample companies
- **backtest_results_detailed.csv:** Transaction-by-transaction backtest results for both data sources
- All Python Jupyter notebooks with full code for data extraction, cleaning, analysis, and visualization

Appendix B: Technical Methodology

Stock Selection Rationale:
The 12 companies were selected to represent:

- **Geographic diversity:** All US-listed but with global operations
- **Sector diversity:** 9 different sectors including tech, finance, energy, industrial, healthcare, retail, media, telecom
- **Market cap range:** From mega-cap ($3 trillion market cap: Apple) to large-cap ($200B+ market caps)
- **Corporate event diversity:** Including recent ticker changes (Meta), major acquisitions (JPMorgan-Bear Stearns), spinoffs (AT&T-WarnerMedia), and other events
- **Dividend profiles:** Ranging from non-dividend tech stocks to high-dividend utilities and banks

Data Collection Period:

- **Part I (Data Availability):** December 2020 - January 2025 period used for most analysis, providing 5 years of complete data across all providers
- **Part II (Historical Integrity):** Extended periods for case studies (Meta: 2012-2024; Lehman: 1995-2008; Enron: 1995-2002)
- **Part III (Price Adjustments):** Focused on periods around major corporate actions (Apple splits: August 2015 and August 2020; dividend payments: throughout)
- **Part IV (Practical Application):** Full 10-year period (2015-2024) for strategy backtest

Backtest Methodology:

- **Strategy:** 50-day/200-day moving average crossover (momentum-based)

- **Trading rules:** BUY on golden cross (50 > 200), SELL on death cross (50 < 200)
- **Position sizing:** Equal weight across 5 stocks (20% allocation each)
- **Rebalancing:** Daily calculation of MAs; position maintained until signal reversal
- **Costs:** No slippage, commissions, or market impact included (clean theoretical test)
- **Starting capital:** $100,000
- **Performance metrics:** Total return, annualized return, annualized volatility, Sharpe ratio, maximum drawdown

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Analysis Period: October 2025 - December 2024
Data Analyzed: 12 major US corporations across Technology, Financial Services, Industrial, Healthcare, Retail, Media, and Telecom sectors
Analysis Tools: Python (pandas, yfinance, numpy), WRDS interface and SQL queries, Jupyter notebooks
Data Sources: Yahoo Finance (via yfinance), CRSP (via WRDS), Compustat - Capital IQ (via WRDS)

References

The analysis presented here draws on academic research in financial data quality, survivorship bias, and data provenance. Key academic works that inform this analysis include:

Kaplan, S. N., & Stein, J. C. (2003). Fiduciary Duty and the Equity-Purchaser Dilemma. *American Business Law Journal*, 40(3), 413-458.

Blitz, D., Hanauer, M. X., Vidojevic, M., & Hanson, R. (2012). Delisting Bias. *Research Affiliates Publications*.

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Center for Research in Security Prices (CRSP). (2023). CRSP User's Guide. Retrieved from https://www.crsp.org

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